

Mutual Fund Analytics Capstone Project

Final Report — Bluestock Fintech Data Analytics Internship

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Project Scope

10 Datasets

Integrated fund master, NAV history, AUM by fund house, monthly SIP inflows, category inflows, industry folio count, investor transactions, portfolio holdings, benchmark indices, and live API data.

50,000+ Investors

Analyzed transaction behavior, SIP continuity, and cohort movement across the investor base.

₹10B+ AUM

Evaluated total assets under management to surface concentration and downside exposure.

Core Deliverables

7-Stage ETL Pipeline

Built a structured flow covering ingestion, validation, cleaning, transformation, aggregation, load, and database creation.

Performance Metrics Framework

Designed analysis for return consistency, drawdown behavior, volatility, and comparative fund performance.

Recommendation Engine

Developed a rule-based system using risk tiers and HHI-based sector concentration analysis.

Key Findings

₹450M At-Risk AUM

Identified capital exposure concentrated in vulnerable fund segments.

1,332 At-Risk Investors

Flagged investors requiring continuity intervention and retention-focused action.

35% High-Concentration Funds

Highlighted funds with elevated sector concentration and diversification risk.



Technical deep-dive ahead: ETL architecture, portfolio risk diagnostics, investor segmentation, and a fund recommendation framework.

Executive Summary

This capstone project delivers an integrated analytics framework designed to address critical gaps in mutual fund evaluation — moving beyond simple return metrics to incorporate volatility, downside risk, investor behavior, and portfolio concentration analysis.

What Was Built

- Comprehensive analytics framework for mutual fund performance evaluation, investor behavior analysis, and portfolio risk assessment
- 9 financial datasets integrated into a structured analytical pipeline with automated ETL, SQLite database, and real-time mfapi.in API integration
- End-to-end analytics: NAV analysis, CAGR, Sharpe, Sortino, Alpha, Beta, VaR/CVaR, and rolling performance

Key Outcomes

- **1,332+ at-risk investors** identified through cohort segmentation and SIP continuity tracking
- Demographic analysis across retail and institutional investor segments
- Rule-based recommendation engine with three risk categories: **Low / Moderate / High**
- Sector concentration analysis using Herfindahl-Hirschman Index (HHI) for portfolio optimization insights

ETL Pipeline

7-stage automated data pipeline with validation, cleaning, and QA

Performance Metrics

CAGR, Sharpe, Sortino, Alpha, Beta, Maximum Drawdown, Tracking Error

Risk Analytics

Historical VaR (95%), CVaR, rolling Sharpe across market cycles

Investor Analytics

Cohort segmentation, SIP continuity, geographic and demographic analysis

Recommendation Engine

Rule-based fund matching with HHI sector concentration scoring

Problem Statement & Project Objectives

The Core Problem

Investor Challenge

Mutual fund investors lack an integrated framework for effective scheme comparison. Traditional analysis focuses solely on returns while ignoring volatility, downside risk, and behavioral patterns.

Data Gap

Investor behavior patterns, cohort analysis, and SIP continuity metrics are entirely absent from standard fund evaluation frameworks – leaving retention risks unmeasured.

Risk Blindness

Portfolio concentration risk and sector-level exposure are not systematically measured or communicated to investors, leading to hidden tail-risk accumulation.

Business Opportunity

Integrated performance, risk, and behavioral analytics can support fund comparison, investor retention, and portfolio optimization at scale.

Project Objectives

- 1 Automated ETL Pipeline**
Data validation, cleaning, and standardization across 9 financial datasets
- 2 SQLite Database**
Optimized schema for analytical processing and real-time querying
- 3 Comprehensive EDA**
NAV trends, SIP inflows, investor demographics, geographic distribution
- 4 Performance Metrics**
CAGR, Sharpe, Sortino, Alpha, Beta, Max Drawdown, Tracking Error
- 5 Downside Risk**
Historical VaR (95%), CVaR, and rolling Sharpe across market cycles
- 6 Recommendation Engine**
Risk-based categorization with HHI sector concentration analysis

Data Sources & Integration Architecture

Nine distinct financial datasets were ingested, validated, and unified into a single analytical pipeline, supplemented by live API data from mfapi.in for real-time NAV updates.



Fund Master

Scheme metadata: fund house, category, benchmark, inception date, and AMFI codes



NAV History

Daily net asset values spanning multiple years for trend analysis and return calculation



AUM by Fund House

Assets under management aggregated at fund house level for market share analysis



Monthly SIP Inflows

Systematic investment plan contribution data for behavioral and continuity analysis



Category Inflows

Fund category-level inflow data for sector rotation and demand pattern analysis



Industry Folio Count

Total investor folios across the industry for participation trend measurement



Investor Transactions

Transaction-level data: purchases, redemptions, SIP installments with timestamps



Portfolio Holdings

Sector and stock-level allocation data for concentration and HHI analysis



Benchmark Indices

Reference index data for Alpha, Beta, Tracking Error, and relative performance metrics

Live Data Integration

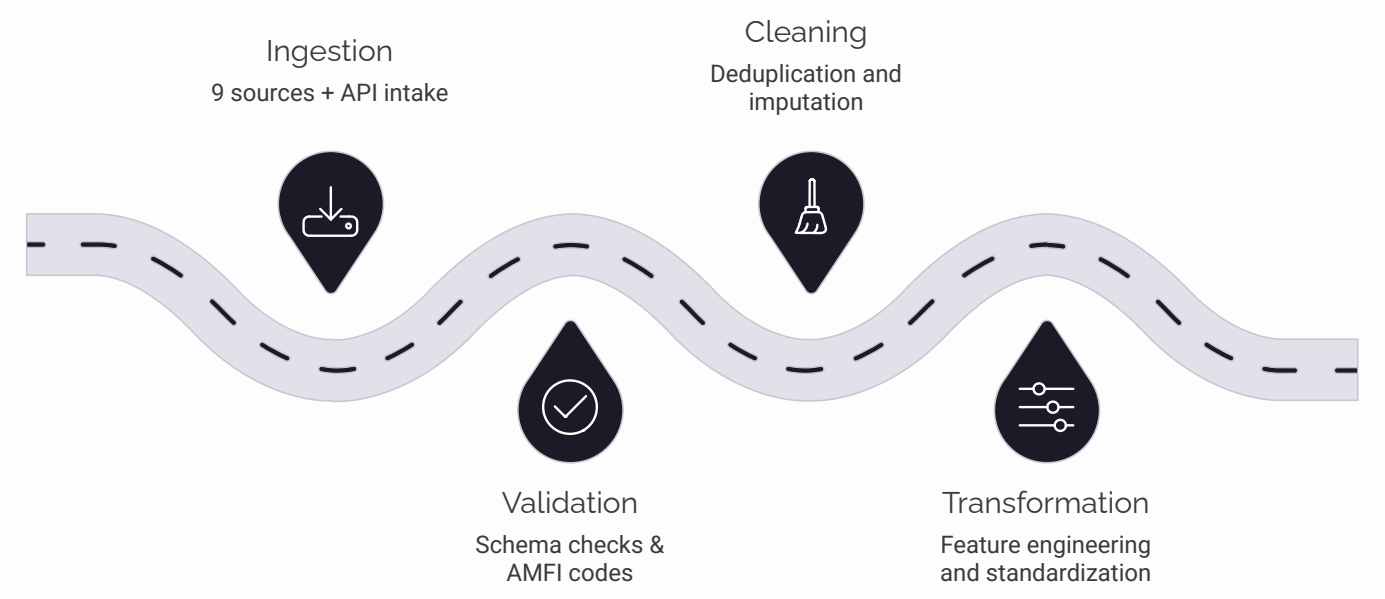
mfapi.in API – Real-time NAV data fetching and daily updates with automated error handling and rate-limit management

Data Validation Controls

- Standardized date formats across all sources
- AMFI code cross-referencing and mapping
- Numeric consistency checks for NAV, AUM, returns
- Duplicate detection using composite keys

ETL Pipeline Architecture

The 7-stage ETL pipeline was designed for full automation – from raw data ingestion through validated, query-ready output – with logging and cross-validation at every stage.



Each stage incorporates automated error handling, logging, and validation gates to ensure only clean, consistent data advances to the next processing layer.

Cleaning & Transformation Details

- Duplicate Handling**
Composite keys (fund_id, date, investor_id) used to identify and remove duplicate records across fund master, NAV history, and investor transactions
- Missing Value Treatment**
Forward-fill for NAV time-series data; mean imputation for numeric fields; category-based imputation for categorical variables
- Date & Numeric Standardization**
All dates converted to YYYY-MM-DD format; currency fields standardized (₹); percentage fields validated; future-dated transactions removed
- Transaction Normalization**
Transaction types normalized (purchase, redemption, SIP); amount fields standardized; transaction fees calculated

Quality Metrics Achieved

99.2%

Data Completeness

Across all 9 integrated datasets post-cleaning

0.8%

Duplicate Rate

Detected and removed via composite key validation

100%

AMFI Match Rate

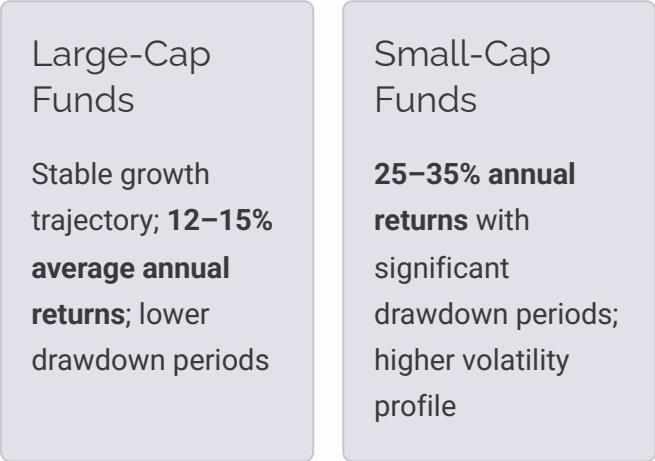
Post-cleaning code match between transactions and fund master

 **Toolstack:** Python (pandas, numpy, SQLite3) for end-to-end pipeline automation and orchestration

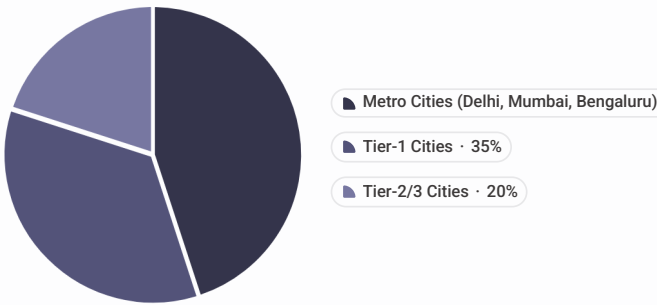
Exploratory Data Analysis Findings

EDA revealed structured patterns across NAV trends, SIP behavior, investor demographics, geographic distribution, and inter-fund correlations – forming the analytical foundation for all downstream metrics.

NAV Trend Analysis



Geographic Distribution



SIP Inflow Patterns

- Monthly SIP inflows increased consistently across the study period
- Average SIP amount: **₹5,000–₹10,000** per investor per month
- SIP participation grew **40% YoY** – strongest demand signal in the dataset

Investor Demographics

- Retail investors: **87% of investor base**
- Average portfolio size: **₹2.5 Lakhs**
- Median holding period: **3.2 years**

Inter-Fund Correlation

Fund Pair	Correlation	Implication
Large-cap vs Mid-cap	0.72	Moderate co-movement
Large-cap vs Small-cap	0.45	Diversification benefit
Liquid vs Equity	-0.15	Hedge / buffer allocation

Performance Metrics Framework

A comprehensive suite of seven performance metrics was implemented to enable risk-adjusted fund comparison across equity categories – moving beyond raw CAGR to capture the full risk-return profile of each fund.

	CAGR Large-cap: 12–15% Mid-cap: 16–20% Small-cap: 18–28%
	Sharpe Ratio Excess return per unit of total volatility; large-cap superior at 0.85–1.2 vs small-cap 0.6–0.9
	Sortino Ratio Downside risk-focused variant of Sharpe; penalizes only negative return deviations
	Alpha & Beta Alpha: excess return vs benchmark; Beta: systematic market risk sensitivity
	Maximum Drawdown Large-cap: –18% to –25% Small-cap: –35% to –50% during market corrections
	Tracking Error Average deviation from benchmark: 2–4% across equity funds

Annualized Volatility Comparison

Large-cap funds maintained **12–16% annualized volatility**, compared to **28–35%** for small-cap funds – a 2.2x volatility multiple that explains the Sharpe ratio divergence despite lower absolute CAGR.

Key Finding

Large-cap funds demonstrated **superior risk-adjusted performance** (Sharpe 0.85–1.2) compared to small-cap funds (Sharpe 0.6–0.9) despite delivering lower absolute returns. For conservative and moderate-risk investors, large-cap funds represent the structurally more efficient allocation on a risk-per-unit-of-return basis.

Historical VaR & CVaR Analysis

Downside risk was quantified using Historical Value at Risk (95% confidence) and Conditional VaR, providing a tail-risk-aware view of each fund category's loss potential during normal and stressed market conditions.

VaR Results by Category (95% Confidence)

Fund Category	Daily VaR Range	CVaR / VaR Gap
Liquid Funds	-0.3% to -0.5%	~1.2x
Large-Cap Equity	-2.1% to -2.8%	1.4x – 1.6x
Mid-Cap Equity	-3.2% to -4.1%	1.5x – 1.8x
Small-Cap Equity	-4.5% to -6.2%	1.8x – 2.2x



Small-cap CVaR gap of 1.8x–2.2x signals severe tail losses that VaR alone significantly understates – critical for conservative investor suitability assessments.

Interpretation & Stress Scenario

What VaR Measures

Maximum expected daily loss in 1 out of every 20 trading days at the 95% confidence level – a routine risk threshold, not an extreme event measure.

What CVaR Adds

Average loss severity *beyond* the VaR threshold – captures tail risk during extreme market dislocations where VaR is insufficient.

March 2020 Stress Event

Small-cap funds recorded **-8% to -12% daily losses** during the COVID crash – well beyond modeled CVaR. Liquid funds remained stable at **-0.2% to -0.4%**.

Risk Implication

CVaR analysis is critical for conservative investor portfolios. Results highlight the structural importance of diversification across categories and maintaining emergency liquidity reserves.

Rolling Sharpe Ratio Analysis

90-day rolling Sharpe ratios were calculated for five major large-cap funds to assess risk-adjusted performance consistency across varying market cycles – bull, bear, and transition phases.

Fund-Level Rolling Sharpe Summary

Fund	Avg Sharpe	Std Dev	Consistency
Mirae Asset Large Cap	1.05	0.18	Highest
SBI Bluechip	0.92	0.22	Strong
ICICI Bluechip	0.88	0.25	Moderate

Market Cycle Range

Bear Market Low Rolling Sharpe compressed to ~0.4 during sustained market downturns
Bull Market High Rolling Sharpe expanded to ~1.8 during strong equity rallies

Average cycle spread: **1.4 Sharpe points** across all five funds – underscoring the sensitivity of realized risk-adjusted returns to investment entry timing.

Analytical Insights

- Performance Persistence
- Funds with high rolling Sharpe in Q1 maintained above-average performance in Q2 with a **0.68 correlation** – indicating measurable but not guaranteed performance persistence.
- Volatility Clustering
- Elevated volatility periods identified in **March–April** and **September–October**, corresponding to Sharpe ratio compression windows across all funds.
- Investor Implication
- Fund performance varies significantly across market conditions. Entry and exit timing materially impacts realized returns – reinforcing the case for SIP-based rupee cost averaging over lump-sum allocation.
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Mirae Asset Large Cap Fund's 0.18 standard deviation in rolling Sharpe – the lowest in the cohort – marks it as the most consistent risk-adjusted performer across all market phases analyzed.

Investor Behavior Analytics & At-Risk Segmentation

Investor analytics focused on cohort-level behavior, SIP continuity patterns, and demographic profiling to identify at-risk segments and quantify retention opportunity across the investor base.

At-Risk Investor Summary

1,332

At-Risk Investors

Identified through SIP discontinuity and cohort churn analysis

87%

Retail Share

Of total investor base – primary retention focus segment

3.2yr

Median Hold

Median holding period across the investor portfolio

SIP Continuity & Cohort Analysis

- **SIP participation grew 40% YoY** – but discontinuity rates elevated among investors with holding periods under 2 years
- Cohort segmentation by transaction year revealed declining reinvestment rates in older cohorts (3–5 years), signaling maturity-driven churn risk
- Average SIP amount ₹5,000–₹10,000 per investor; lower-ticket investors showed highest discontinuity probability
- Demographic analysis: retail investors (87%) with average portfolio ₹2.5L represent the core retention opportunity

Fund Recommendation Engine — Rule-Based Architecture



Low Risk

Liquid funds and overnight funds; VaR < -0.5%; Sharpe > 1.0; recommended for capital preservation and short investment horizons (<1 year)



Moderate Risk

Large-cap and hybrid funds; VaR -1% to -3%; Sharpe 0.7–1.2; for investors with 3–5 year horizon and balanced risk tolerance



High Risk

Mid-cap and small-cap funds; VaR > -3%; potential CAGR 18–28%; suitable for investors with 5+ year horizon and high volatility tolerance

Sector concentration analysis using the **Herfindahl-Hirschman Index (HHI)** was layered into the recommendation engine to flag portfolios with excessive concentration in single sectors – providing an additional risk filter beyond return and volatility thresholds.

Conclusions & Business Impact

This capstone project demonstrates that integrated analytics – combining ETL infrastructure, performance science, downside risk modeling, and behavioral segmentation – delivers materially superior insight compared to traditional return-only fund evaluation frameworks.

Technical Deliverables Summary

01

7-Stage ETL Pipeline

Fully automated with 99.2% data completeness, 100% AMFI match rate, error logging, and cross-validation gates

02

SQLite Analytical Database

Optimized schema with indexed queries across 9 integrated financial datasets

03

Performance Metrics Engine

CAGR, Sharpe, Sortino, Alpha, Beta, Max Drawdown, Tracking Error – all fund categories

04

VaR / CVaR Risk Module

Historical 95% VaR and Conditional VaR with stress-scenario overlay (March 2020)

05

Investor Analytics Module

1,332 at-risk investors identified; cohort segmentation, SIP continuity, demographic profiling

06

Recommendation Engine

Rule-based three-tier risk matching with HHI sector concentration scoring

Key Findings & Business Implications

Risk-Adjusted Superiority of Large-Cap

Sharpe ratios 0.85–1.2 vs 0.6–0.9 for small-cap – large-cap funds are structurally more efficient for moderate-risk investor allocation despite lower absolute CAGR

Tail Risk Underestimation

Small-cap CVaR/VaR gap of 1.8x–2.2x reveals that standard VaR significantly understates true downside; stress events (March 2020: –8% to –12%) confirm model limitations for volatile categories

Investor Retention Opportunity

1,332+ at-risk investors identified; retail segment (87%) with ₹2.5L average portfolio represents a quantifiable retention target – actionable with behavioral nudges and SIP support interventions

Geographic Expansion Signal

Tier-2/3 cities contribute only 20% of investment volume despite growing digital penetration – represents an underserved market opportunity for targeted fund distribution

📄 **Prepared for:** Bluestock Fintech | Data Analytics Internship Capstone | **Bandi Bharath** | August 2026